

Dimension Reduction Using PCA

CSE427: Machine Learning

Why Do We Need Dimension Reduction?

Many real datasets have many features.

$$x_1, x_2, x_3, \dots, x_d$$

Examples:

- image data: thousands of pixels,
- text data: thousands of word features,
- gene expression data: thousands of genes,
- student data: attendance, study hour, assignment, quiz, lab score.

Problem

High-dimensional data can be expensive, noisy, redundant, and difficult to visualize.

Goal

Represent the data using fewer features while keeping the most important information.

What is PCA?

PCA means:

Principal Component Analysis

Definition

PCA is an unsupervised dimensionality reduction method that finds new axes where the data has maximum variance.

Original features:

$$x_1, x_2, \dots, x_d$$

New PCA features:

$$PC_1, PC_2, \dots, PC_k$$

where:

$$k < d$$

PCA Does Not Delete Features

Suppose we have two original features:

$$x_1 = \text{Study Hours}$$

$$x_2 = \text{Practice Problems}$$

PCA may create:

$$PC_1 = 0.71x_1 + 0.70x_2$$

This means PC_1 combines both features.

Interpretation

If study hours and practice problems increase together, then PC_1 can represent overall preparation.

Common Mistake

PCA is not feature selection. Feature selection keeps original features. PCA creates transformed features.

Why Variance Matters in PCA

PCA tries to preserve directions where data changes the most.

Large variance $\Rightarrow$ more information

Small variance $\Rightarrow$ less information or noise

Main PCA Question

Which direction captures the maximum spread of the data?

The first principal component is:

PC_1 = direction of maximum variance

The second principal component is:

PC_2 = next maximum variance direction perpendicular to PC_1

Geometric Intuition of PCA

In 2D, data may spread mostly along a diagonal direction.

Instead of representing each point using:

$$(x_1, x_2)$$

we can represent it by its position along the best direction:

$$z = x \cdot PC_1$$

Meaning

Projection onto PC_1 gives a lower-dimensional representation.

Original representation:

$$\mathbb{R}^2$$

Reduced representation:

$$\mathbb{R}^1$$

PCA Algorithm

Given a data matrix:

$$X \in \mathbb{R}^{n \times d}$$

where:

n = number of samples

d = number of features

PCA steps:

- 1 Center the data.
- 2 Compute covariance matrix.
- 3 Find eigenvalues and eigenvectors.
- 4 Sort eigenvectors by eigenvalues.
- 5 Select top k eigenvectors.
- 6 Project data onto selected eigenvectors.

Why Do We Center the Data?

PCA studies variation around the mean.

For each feature:

$$x'_j = x_j - \bar{x}_j$$

where:

$$\bar{x}_j = \text{mean of feature } j$$

Why Centering is Necessary

Without centering, PCA may capture the location of the data instead of the direction of variation.

After centering:

$$\text{mean of each feature} = 0$$

What is Covariance?

Covariance measures how two features move together.

$$\text{Cov}(x_1, x_2) = \frac{1}{n-1} \sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$$

Covariance	Meaning
Positive	Features increase together
Negative	One increases, other decreases
Near zero	Weak linear relationship

PCA Uses Covariance

PCA uses covariance to understand feature relationships and data spread.

Covariance Matrix

For two features:

$$C = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix}$$

Matrix form:

$$C = \frac{1}{n-1} X_c^T X_c$$

where X_c is centered data.

Meaning

The covariance matrix tells PCA how the data is spread in different directions.

Eigenvalues and Eigenvectors in PCA

PCA solves:

$$Cv = \lambda v$$

where:

C = covariance matrix

v = eigenvector

λ = eigenvalue

Meaning

Eigenvectors give principal component directions. Eigenvalues tell how much variance exists in those directions.

Large $\lambda \Rightarrow$ important principal component

Meaningful Example: Student Preparation Data

Suppose we have two features:

$x_1 = \text{Study Hours}$

$x_2 = \text{Practice Problems Solved}$

Student	Study Hours x_1	Practice Problems x_2
A	2	3
B	3	4
C	4	6
D	5	7
E	6	9

Goal

Reduce two correlated features into one principal component representing preparation level.

Step 1: Compute Feature Means

Mean of study hours:

$$\bar{x}_1 = \frac{2+3+4+5+6}{5}$$

$$\bar{x}_1 = \frac{20}{5} = 4$$

Mean of practice problems:

$$\bar{x}_2 = \frac{3+4+6+7+9}{5}$$

$$\bar{x}_2 = \frac{29}{5} = 5.8$$

So:

$\bar{x}_1 = 4, \quad \bar{x}_2 = 5.8$
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Step 2: Center the Data

$$x'_1 = x_1 - 4$$

$$x'_2 = x_2 - 5.8$$

Student	x_1	x_2	x'_1	x'_2
A	2	3	-2	-2.8
B	3	4	-1	-1.8
C	4	6	0	0.2
D	5	7	1	1.2
E	6	9	2	3.2

$$X_c = \begin{bmatrix} -2 & -2.8 \\ -1 & -1.8 \\ 0 & 0.2 \\ 1 & 1.2 \\ 2 & 3.2 \end{bmatrix}$$

Step 3: Compute Covariance Matrix

$$C = \frac{1}{n-1} X_c^T X_c$$

Here:

$$n = 5$$

First compute:

$$\text{Var}(x_1) = \frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{4}$$

$$\text{Var}(x_1) = \frac{4 + 1 + 0 + 1 + 4}{4} = \frac{10}{4} = 2.5$$

$$\text{Var}(x_2) = \frac{(-2.8)^2 + (-1.8)^2 + (0.2)^2 + (1.2)^2 + (3.2)^2}{4}$$

$$\text{Var}(x_2) = \frac{7.84 + 3.24 + 0.04 + 1.44 + 10.24}{4} = \frac{22.8}{4} = 5.7$$

Step 4: Compute Covariance

$$\text{Cov}(x_1, x_2) = \frac{\sum x_1' x_2'}{n - 1}$$

$$\sum x_1' x_2' = (-2)(-2.8) + (-1)(-1.8) + (0)(0.2) + (1)(1.2) + (2)(3.2)$$

$$= 5.6 + 1.8 + 0 + 1.2 + 6.4$$

$$= 15$$

Therefore:

$$\text{Cov}(x_1, x_2) = \frac{15}{4} = 3.75$$

So the covariance matrix is:

$$C = \begin{bmatrix} 2.5 & 3.75 \\ 3.75 & 5.7 \end{bmatrix}$$

Step 5: Find Eigenvalues

Solve:

$$|C - \lambda I| = 0$$

$$C - \lambda I = \begin{bmatrix} 2.5 - \lambda & 3.75 \\ 3.75 & 5.7 - \lambda \end{bmatrix}$$

$$(2.5 - \lambda)(5.7 - \lambda) - (3.75)^2 = 0$$

$$(2.5 - \lambda)(5.7 - \lambda) - 14.0625 = 0$$

$$14.25 - 8.2\lambda + \lambda^2 - 14.0625 = 0$$

$$\lambda^2 - 8.2\lambda + 0.1875 = 0$$

Step 6: Solve the Eigenvalue Equation

$$\lambda^2 - 8.2\lambda + 0.1875 = 0$$

Using quadratic formula:

$$\lambda = \frac{8.2 \pm \sqrt{(8.2)^2 - 4(1)(0.1875)}}{2}$$

$$\lambda = \frac{8.2 \pm \sqrt{67.24 - 0.75}}{2}$$

$$\lambda = \frac{8.2 \pm \sqrt{66.49}}{2}$$

$$\sqrt{66.49} \approx 8.154$$

Therefore:

$$\lambda_1 = \frac{8.2 + 8.154}{2} = 8.177$$

$$\lambda_2 = \frac{8.2 - 8.154}{2} = 0.023$$

$$\lambda_1 = 8.177, \quad \lambda_2 = 0.023$$

Step 7: Interpretation of Eigenvalues

Total variance:

$$\lambda_1 + \lambda_2 = 8.177 + 0.023 = 8.2$$

Variance explained by PC_1 :

$$\frac{8.177}{8.2} = 0.9972$$

$$= 99.72\%$$

Variance explained by PC_2 :

$$\frac{0.023}{8.2} = 0.0028$$

$$= 0.28\%$$

Conclusion

Almost all variance is captured by PC_1 . So reducing from 2D to 1D is reasonable.

Step 8: Eigenvector for PC_1

Use:

$$(C - \lambda_1 I)v = 0$$

$$\lambda_1 = 8.177$$

$$\begin{bmatrix} 2.5 - 8.177 & 3.75 \\ 3.75 & 5.7 - 8.177 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -5.677 & 3.75 \\ 3.75 & -2.477 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

From the first row:

$$-5.677v_1 + 3.75v_2 = 0$$

$$3.75v_2 = 5.677v_1$$

Step 9: Normalize the Eigenvector

Unnormalized eigenvector:

$$v = \begin{bmatrix} 1 \\ 1.514 \end{bmatrix}$$

Magnitude:

$$\|v\| = \sqrt{1^2 + (1.514)^2}$$

$$\|v\| = \sqrt{1 + 2.292}$$

$$\|v\| = \sqrt{3.292} = 1.814$$

Normalized eigenvector:

$$PC_1 = \frac{1}{1.814} \begin{bmatrix} 1 \\ 1.514 \end{bmatrix}$$

$$PC_1 = \begin{bmatrix} 0.551 \\ 0.835 \end{bmatrix}$$

Meaning

Practice problems receive slightly higher weight than study hours in this principal direction.

Step 10: Project Data onto PC_1

Projection formula:

$$z_i = x_i' \cdot PC_1$$

$$PC_1 = \begin{bmatrix} 0.551 \\ 0.835 \end{bmatrix}$$

For student A:

$$z_A = (-2)(0.551) + (-2.8)(0.835)$$

$$z_A = -1.102 - 2.338 = -3.440$$

For student B:

$$z_B = (-1)(0.551) + (-1.8)(0.835)$$

$$z_B = -0.551 - 1.503 = -2.054$$

For student C:

Step 11: Project Remaining Points

For student D:

$$z_D = (1)(0.551) + (1.2)(0.835)$$

$$z_D = 0.551 + 1.002 = 1.553$$

For student E:

$$z_E = (2)(0.551) + (3.2)(0.835)$$

$$z_E = 1.102 + 2.672 = 3.774$$

So the reduced 1D data is:

$$Z = \begin{bmatrix} -3.440 \\ -2.054 \\ 0.167 \\ 1.553 \\ 3.774 \end{bmatrix}$$

Final Reduced Dataset

Original dataset had two features:

$$x_1 = \text{Study Hours}, \quad x_2 = \text{Practice Problems}$$

After PCA, we represent each student using one value:

PC_1 = overall preparation score

Student	PCA Reduced Value
A	-3.440
B	-2.054
C	0.167
D	1.553
E	3.774

Interpretation

Students with higher PCA scores generally studied more and solved more problems.

Why This Example is Meaningful

The two features are correlated:

more study hours $\Rightarrow$ more practice problems

So PCA finds a combined direction:

$$PC_1 = 0.551(\text{Study Hours}) + 0.835(\text{Practice Problems})$$

Interpretation

Instead of two separate features, PCA creates one preparation score.

This reduces:

$$2D \rightarrow 1D$$

while preserving:

99.72%

of the variance.

What Happens to PC_2 ?

Since:

$$\lambda_2 = 0.023$$

variance explained by PC_2 is:

$$0.28\%$$

So PC_2 contains very little information.

Decision

We drop PC_2 because it explains very little variance.

Important

Dropping PC_2 means losing a small amount of information.

How Many Components Should We Keep?

In real datasets, we may have many components:

$$PC_1, PC_2, \dots, PC_d$$

We choose k based on cumulative explained variance:

$$\frac{\lambda_1 + \lambda_2 + \dots + \lambda_k}{\lambda_1 + \lambda_2 + \dots + \lambda_d}$$

Common choices:

$$90\%, \quad 95\%, \quad 99\%$$

Example

If first 10 PCs explain 95% variance, we may reduce 100 features to 10 features.

PCA vs Feature Selection

Topic	PCA	Feature Selection
Output	New transformed features	Original features
Uses labels?	No	Sometimes yes
Interpretability	Lower	Higher
Goal	Preserve variance	Keep important features
Example	$PC_1 = 0.55x_1 + 0.84x_2$	Keep x_2 , drop x_1

Important

PCA is unsupervised. It does not know the class label or target value.

When PCA is Useful

PCA is useful for:

- visualization,
- noise reduction,
- compression,
- speeding up machine learning algorithms,
- removing redundancy among correlated features.

Example:

1000 features $\rightarrow$ 50 principal components

Main Benefit

The model trains on fewer variables while keeping most of the data variation.

Limitations of PCA

PCA also has limitations:

- PCA is linear.
- PCA is sensitive to feature scale.
- PCA may reduce interpretability.
- PCA preserves variance, not necessarily class separation.
- PCA can be affected by outliers.

Important

High variance does not always mean high predictive importance.

Example:

A feature may have low variance but still be very useful for classification.

Why Standardization May Be Needed

If features have different units or scales, PCA can be dominated by large-scale features.

Example:

$x_1 = \text{Age in years}$

$x_2 = \text{Income in taka}$

Income has much larger numerical values.

Solution

Standardize each feature:

$$z = \frac{x - \mu}{\sigma}$$

After standardization:

mean = 0, standard deviation = 1

PCA Summary

- PCA reduces dimensionality by creating new features.
- These new features are called principal components.
- PC_1 captures maximum variance.
- PC_2 captures next highest variance and is perpendicular to PC_1 .
- Eigenvectors give component directions.
- Eigenvalues give explained variance.
- Data is projected onto selected components.

$$PCA = Center + Covariance + Eigenvectors + Projection$$

Common Mistakes

- 1 Forgetting to center or standardize data.
- 2 Thinking PCA uses class labels.
- 3 Thinking PCA selects original features.
- 4 Choosing components with smallest eigenvalues.
- 5 Forgetting eigenvectors must be normalized.
- 6 Thinking high variance always means high predictive value.

Most Important

PCA preserves variance, not necessarily classification accuracy.